# SIMATS ENGINEERING
## ASSESSMENT TOOL 1

**COURSE NAME: COMPUTER NETWORKS**
**COURSE CODE: CSA0716**

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**COURSE OUTCOME COVERED**

***CO5:***
Design network topologies star, mesh, bus, and hybrid using networking devices, transmission media, and cabling standards *(BL2, BL5)*

*Assessment Tool Weightage: 40%*

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**QUESTIONS: 3** **Total Marks:30**

Annexure A

## Pseudocode Writing Task (Algorithm Design)

1. In a network simulation using Cisco Packet Tracer, a student records total data received as 8000 bits in 4 seconds and observes that 200 packets were sent while 180 packets were received. Write a pseudocode algorithm to calculate the network throughput in bits per second and the packet loss percentage based on the given values. The algorithm should include validation to ensure that time and total packets sent are not zero to avoid errors. It should display both the throughput and packet loss results clearly. Add logic to classify the network as Efficient if throughput is greater than 1500 bps and packet loss is less than 10%, otherwise classify it as Inefficient. Finally, ensure the algorithm outputs a complete summary of the network performance based on the calculated metrics.

2. An organization has multiple network links connecting its branches. The administrator wants to monitor bandwidth utilization and identify overloaded links. Write a pseudocode algorithm that periodically collects the bandwidth usage of each network link and stores the values in an array or list. Use a loop to continuously monitor all links. Add conditions to detect when bandwidth utilization exceeds 80% and recommends switching traffic to a backup link. Explain how this algorithm supports efficient bandwidth management and prevents network congestion.

3. A company has six branch offices connected through multiple network links. Each link has different bandwidth, delay, and utilization values, which change throughout the day. The network administrator wants to continuously identify the best communication path between the headquarters and each branch based on overall link quality rather than just the shortest distance. Write a pseudocode algorithm that periodically collects the bandwidth, delay, and utilization of every network link and stores the values in suitable arrays or lists. Use a loop to repeat the monitoring process at regular intervals.

Calculate a link quality score for each path using the collected metrics and select the path with the highest score. Include conditions to detect when a link becomes overloaded or fails, automatically selecting an alternate path and generating an alert for the administrator. Explain how your algorithm improves network reliability, routing efficiency, and fault tolerance while adapting to changing network conditions.

**1. Network Throughput and Packet Loss.**

**Given Data:**

- Total Data Received = 8000 bits
- Time = 4 seconds
- Packets Sent = 200
- Packets Received = 180

**Calculation:**

Throughput = 8000 / 4 = 2000 bps
Packet Loss = ((200 - 180) / 200) × 100 = 10%

**Pseudocode:**

```
ALGORITHM Calculate Network Performance
BEGIN
   total Data ← 8000
   time ← 4
   sent ← 200
   received ← 180
   IF time = 0 OR sent = 0 THEN
      DISPLAY "Invalid Input"
      STOP
   END IF
   throughput ← total Data / time
   packet Loss ← ((sent - received) / sent) × 100
   IF throughput > 1500 AND packet Loss < 10 THEN
      result ← "Efficient"
   ELSE
      result ← "Inefficient"
   END IF
   DISPLAY throughput
   DISPLAY packet Loss
   DISPLAY result
END
```

**Output:**

Throughput = 2000 bps
Packet Loss = 10%
Network Classification = Inefficient

**2. Bandwidth Utilization Monitoring and Overloaded Link Detection.**

**Introduction:**
This algorithm continuously monitors multiple network links and recommends switching traffic to a backup link when utilization exceeds 80%.

**Pseudocode:**
```
ALGORITHM Monitor Bandwidth
BEGIN
   links ← [Link1, Link2, Link3, Link4]
   WHILE TRUE DO
      FOR each link IN links DO
         usage ← GET BANDWIDTH (link)
         IF usage > 80 THEN
            DISPLAY "Link Overloaded"
            SWITCH traffic to Backup Link
         ELSE
            DISPLAY "Link Working Normally"
         END IF
      END FOR
   END WHILE
END
```

**Advantages:**
- Continuously monitors network links.
- Detects overloaded links.
- Switches traffic to backup link.
- Prevents congestion.
- Improves reliability.

**3. Best Communication Path Selection.**

**Introduction:**
The algorithm collects bandwidth, delay, utilization, and link status to select the best communication path.

**Quality Score:**
Quality Score = (0.5 × Bandwidth Score) + (0.3 × Delay Score) + (0.2 × Utilization Score)

**Pseudocode:**
```
ALGORITHM Select Best Path
BEGIN
   WHILE TRUE DO
      Collect network metrics
      Calculate quality score
      Select highest score path
      IF link fails OR utilization > 80 THEN
         Select alternate path
         Generate alert
      END IF
   END WHILE
END
```

**Advantages:**

- Improves reliability.
- Selects best path.
- Supports fault tolerance.
- Avoids overloaded links.
- Adapts to changing conditions.

## RUBRICS FOR CALCULATION

| Criteria | Weightage (%) | Exemplary (4) | Proficient (3) | Developing (2) | Beginning (1) |
|---|---|---|---|---|---|
| **Understanding of Problem & Constraints** | 20% | Clearly identifies all constraints and fully understands the problem requirements | Identifies most constraints with minor gaps in understanding | Identifies some constraints but misses key aspects | Shows little or no understanding of constraints or problem |
| **Problem Decomposition** | 15% | Breaks problem into logical, well-structured sub-problems effectively | Breaks problem into sub-parts with some logical flow | Attempts decomposition but lacks clarity or structure | No clear decomposition of the problem |
| **Application of Constraints in Solution** | 20% | Applies all constraints correctly and consistently in solution design | Applies most constraints correctly with minor errors | Applies some constraints but with noticeable errors | Fails to apply constraints appropriately |
| **Logical Reasoning & Strategy** | 15% | Uses strong, clear, and efficient reasoning strategies | Uses generally sound reasoning with minor gaps | Reasoning is partially correct but inconsistent | Reasoning is unclear or incorrect |
| **Solution Accuracy & Feasibility** | 15% | Solution is fully correct, feasible, and optimized | Solution is mostly correct and feasible | Solution has errors or is partially feasible | Solution is incorrect or not feasible |
| **Creativity & Innovation** | 5% | Demonstrates highly innovative and unique approach | Shows some creativity in approach | Limited creativity shown | No creativity; relies on basic or copied ideas |
| **Clarity of Presentation** | 10% | Solution is clearly presented with excellent organization and explanation | Presentation is clear with minor issues | Presentation lacks clarity or organization | Poorly presented and difficult to understand |